HEALTHCARE DATA ANALYTICS REPORT

Hospital 30-Day Readmission Risk Analysis Using Python

A Healthcare Data Analytics Project

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Dataset: Synthetic sample of 500,000 hospital patients | Tools: Python & Microsoft Excel

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Executive Summary

Hospitals frequently face recurrent admissions, where patients return to hospital within 30 days of being discharged. Frequent 30-day readmissions increase healthcare costs, occupy beds that could be used for other patients, raise the workload of hospital staff, and can point to gaps in discharge planning and follow-up care.

This project analyzed a synthetic dataset of 500,000 hospital patients, created with Claude AI, to identify the factors associated with 30-day readmission. The dataset contains 13 factors: age, gender, race, admission type, length of stay, primary diagnosis, number of previous admissions, number of medications, comorbidity count, discharge disposition, haemoglobin level, creatinine level and insurance type. Using Python, the data were cleaned, missing values were imputed, and descriptive statistics, chi-square tests, independent samples t-tests, the Mann–Whitney U test and logistic regression (with model evaluation) were performed.

Overall, 145,413 patients (29.08%) were readmitted within 30 days. Age, admission type, length of stay, primary diagnosis, previous admissions, comorbidity count, number of medications and discharge disposition showed statistically significant associations with readmission, while gender, race, haemoglobin level, creatinine level and insurance type did not. Age showed the strongest association: readmission rose from 18.34% in patients aged 18–24 to 31.83% in those aged 65 and above. Because the sample is very large, most effects were small, and the logistic regression model could only slightly separate readmitted from non-readmitted patients (AUC = 0.55). The data are synthetic and the results show associations, not causes, so the findings should be treated as an analytical demonstration rather than clinical evidence.

Key Findings at a Glance

500,000 Patients Analyzed	**29.08%** Overall 30-Day Readmission Rate	**Age** Strongest Factor Associated with Readmission	**31.83%** Readmission Rate, Ages 65+

Factor	Significant?	Effect on Readmission	Recommended Action
Age	Yes ($p < 0.001$)	31.83% in ages 65+ vs 18.34% in ages 18–24	Targeted discharge planning for older patients
Gender	No ($p = 0.606$)	Rates almost identical across groups	No targeted action needed
Race	No ($p = 0.132$)	Rates almost identical across groups	No targeted action needed
Admission Type	Yes ($p < 0.001$)	Emergency highest (29.79%); very weak effect	Monitor emergency admissions
Length of Stay	Yes ($p < 0.001$)	Readmitted patients stayed 0.73 days longer (very small effect)	Review discharge-readiness criteria
Primary Diagnosis	Yes ($p < 0.001$)	COPD, sepsis, acute MI and heart failure about 32% vs about 27% for others	Diagnosis-specific discharge pathways
Previous Admissions	Yes ($p < 0.001$)	12.4% higher odds per extra admission	Case management for frequently admitted patients

Factor	Significant?	Effect on Readmission	Recommended Action
Number of Medications	Yes (p < 0.001)	0.34% higher odds per extra medication (tiny)	No targeted action needed
Comorbidity Count	Yes (p < 0.001)	8.2% higher odds per extra condition	Flag patients with several conditions
Discharge Disposition	Yes (p < 0.001)	Against medical advice 34.97%; nursing facility 30.18%	Structured follow-up for these discharges
Haemoglobin	No (p = 0.180)	Averages 12.49 vs 12.50 g/dL	No targeted action needed
Creatinine	No (p = 0.845)	Averages 1.0123 vs 1.0127 mg/dL	No targeted action needed
Insurance Type	No (p = 0.543)	Rates almost identical across groups	No targeted action needed

Full details and additional recommendations are provided in the Limitations, Recommendations and Next Steps sections of this report.

Business Problem

Hospitals aim to reduce 30-day readmissions because they:

- Increase healthcare costs and resource use.
- Occupy beds that could be used for other patients.
- Increase the workload of healthcare staff.
- May indicate gaps in discharge planning, care coordination, or post-discharge follow-up.

The hospital wants to identify the factors associated with 30-day readmission so that management can better identify higher-risk patient groups, improve discharge and follow-up processes, and allocate resources more effectively.

Project Objective

To identify which patient characteristics (Age, Gender, Race, Admission Type, Length of Stay, Primary Diagnosis, Number of Previous Admissions, Number of Medications, Comorbidity Count, Discharge Disposition, Haemoglobin Level, Creatinine Level, and Insurance Type) are associated with 30-day hospital readmission, and to evaluate how well the count variables (previous admissions, medications and comorbidities) can predict readmission.

Dataset and Data Source

The dataset used in this project is synthetic. It was generated with Claude AI (Anthropic) from a written prompt describing the variables listed below, so it contains no real patient information. It has 500,000 rows (one row per patient record) and 14 variables: 13 factors and the outcome, Readmitted_30_Days (Yes/No).

Variable	Type	Description
Readmitted_30_Days	Outcome	Readmitted within 30 days of discharge (Yes/No)
Age	Numerical	Age in years (18–100)
Gender	Categorical	Female, Male, Other, Unknown
Race	Categorical	White, Black, Hispanic, Asian, Other, Unknown
Admission_Type	Categorical	Emergency, Urgent, Elective
LOS_Days	Numerical	Length of stay in days
Primary_Diagnosis	Categorical	15 primary diagnoses
Previous_Admissions	Numerical	Number of previous admissions (0–10)
Num_Medications	Numerical	Number of medications (0–28)
Comorbidity_Count	Numerical	Number of coexisting conditions (0–12)
Discharge_Disposition	Categorical	Home, Skilled Nursing Facility, Rehab Facility, Home Health, Against Medical Advice, Hospice, Expired
Haemoglobin_g_dl	Numerical	Haemoglobin level (g/dL)
Creatinine_mg_dl	Numerical	Creatinine level (mg/dL)
Insurance_Type	Categorical	Medicare, Private, Medicaid, Self-Pay, Other, Unknown

Because the records are synthetic, the findings in this report describe patterns within this dataset and are not clinical evidence.

Methodology

This project followed a structured healthcare data analytics workflow.

1. Data Preparation

- Imported the dataset (Excel workbook with 500,000 rows) into Python and inspected its structure and data types.
- Removed one blank duplicate column and the Patient_ID column, which is only a row label.
- Checked for duplicate records (none found) and for unwanted spaces and inconsistent capitalization (none found).
- Used the IQR method to identify outliers in the seven numerical variables. All flagged values were retained because there was no evidence that they were errors.
- Treated missing values as shown below, so that all 500,000 records were kept.
- Created Age_Group (18–24, 25–34, 35–44, 45–64, 65+) from Age. Final dataset: 500,000 rows × 14 columns, with no missing values.

Variable	Missing Values	Method Used
Age	5.99% (29,965)	Mean imputation (67.42 years)
Gender	4.02% (20,117)	Labelled “Unknown”
Race	2.94% (14,716)	Labelled “Unknown”
Haemoglobin	11.98% (59,912)	Mean imputation (12.50 g/dL)
Creatinine	9.97% (49,837)	Median imputation (0.90 mg/dL)
Insurance Type	2.03% (10,168)	Labelled “Unknown”

Mean imputation was used for Age and Haemoglobin because their distributions are approximately symmetric. Median imputation was used for Creatinine because its distribution is positively skewed. Missing Gender, Race and Insurance Type values were labelled “Unknown” to avoid assuming unavailable information.

2. Descriptive Statistics

- Calculated frequencies, percentages, means, medians, standard deviations, quartiles, skewness and kurtosis.
- Summarized patient characteristics using tables and charts.

3. Chi-Square Test of Independence

Compared readmission rates across the categorical factors (age group, gender, race, admission type, primary diagnosis, discharge disposition and insurance type). Cramér's V was used to measure the strength of each association.

4. Independent Samples t-Test and Mann–Whitney U Test

Compared the average age, length of stay, haemoglobin level and creatinine level between readmitted and non-readmitted patients. Cohen's d was used to measure the size of each difference. Because length of stay is strongly skewed, the Mann–Whitney U test was added as a check, and Welch's t-test (which does not assume equal variances) was used to confirm the t-test results.

5. Logistic Regression

Logistic regression was performed for the three count variables (number of previous admissions, number of medications and comorbidity count) to estimate odds ratios for 30-day readmission. McFadden's pseudo-R^2 and the likelihood-ratio test were used to assess the model.

6. Model Evaluation

The patients were randomly split into a training set (80%, 400,000 patients) and a test set (20%, 100,000 patients). The model was fitted on the training set and evaluated on the unseen test set using the AUC, a confusion matrix, sensitivity, specificity, precision and calibration.

7. Hypothesis Testing

- For each factor, the null hypothesis was that it is not associated with 30-day readmission.
- Statistical significance was assessed using a 5% significance level ($\alpha = 0.05$).
- Decisions were based on p-values together with effect sizes (Cramér's V, Cohen's d, odds ratios).

Skills & Tools

Software

- Python (pandas, NumPy, SciPy, scikit-learn, Matplotlib)
- Microsoft Excel (source workbook)

Functions Used

Function	Function	Function
read_excel()	isna()	duplicated()
fillna()	groupby()	crosstab()
cut()	chi2_contingency()	ttest_ind()
mannwhitneyu()	train_test_split()	LogisticRegression()
roc_auc_score()	confusion_matrix()	brier_score_loss()

Statistical Techniques

- Descriptive Statistics
- Chi-Square Test of Independence and Cramér's V
- Independent Samples t-Test and Cohen's d
- Mann–Whitney U Test
- Logistic Regression and Odds Ratios
- Model Evaluation (AUC, Confusion Matrix, Calibration)
- Hypothesis Testing

Applied & Analytical Skills

- Analytical Skills
- Data Cleaning and Missing-Data Imputation
- Data Transformation
- Data Visualization
- Healthcare Data Analysis
- Statistical Reporting
- Business Insight Generation

Dataset Overview

Before examining individual factors, the composition of the 500,000-patient dataset was reviewed. Figures 1 to 3 show the categorical variables and Table 4 summarizes the numerical variables. Percentages are rounded and may not add to exactly 100%.

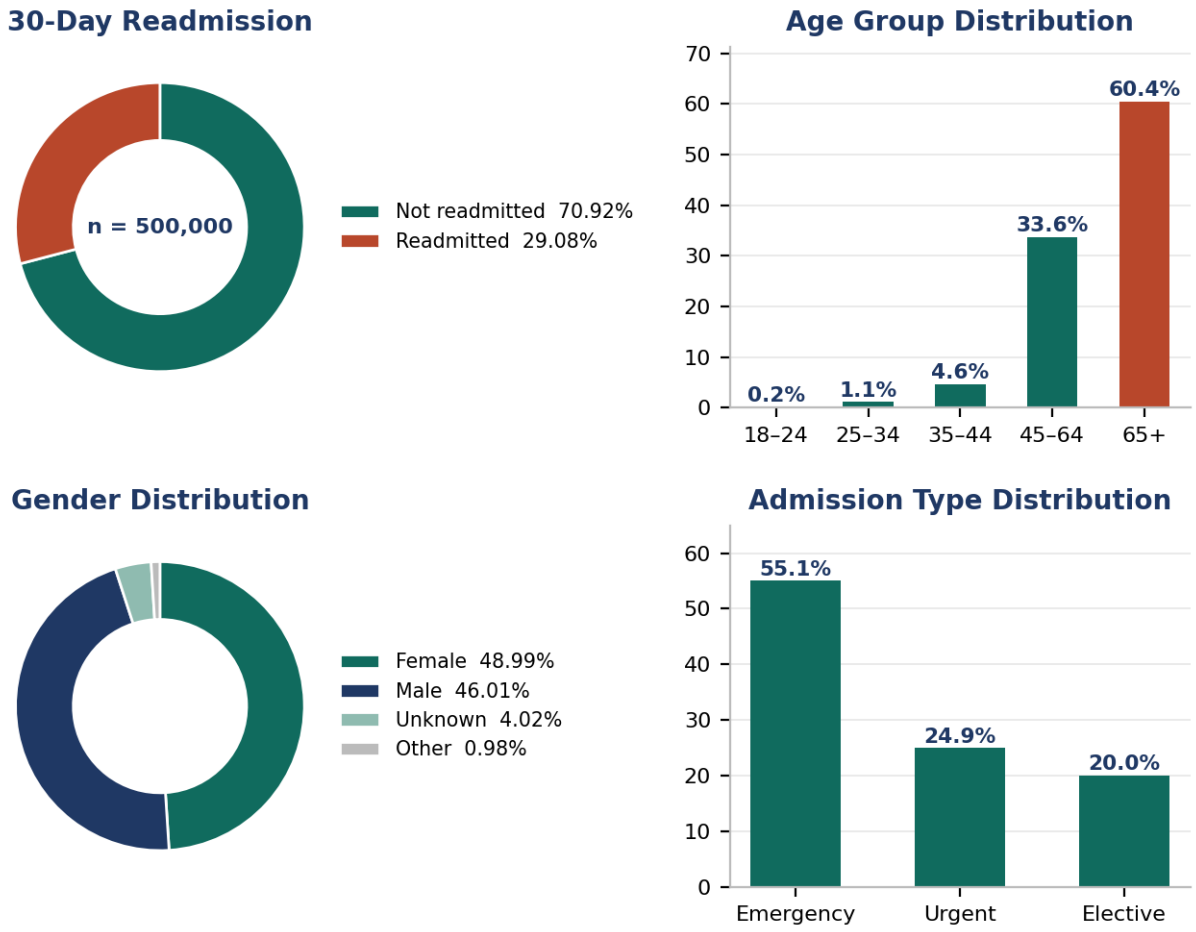


Figure 1. Dataset overview: 30-day readmission, age group, gender and admission type.

145,413 patients (29.08%) were readmitted within 30 days and 354,587 (70.92%) were not. The dataset is made up mainly of older patients: 60.44% are aged 65 and above and 33.64% are aged 45–64. The gender split is close to even (female 48.99%, male 46.01%; 4.02% unknown and 0.98% other). Emergency admissions form the largest group (55.06%), followed by urgent (24.94%) and elective (20.01%) admissions.

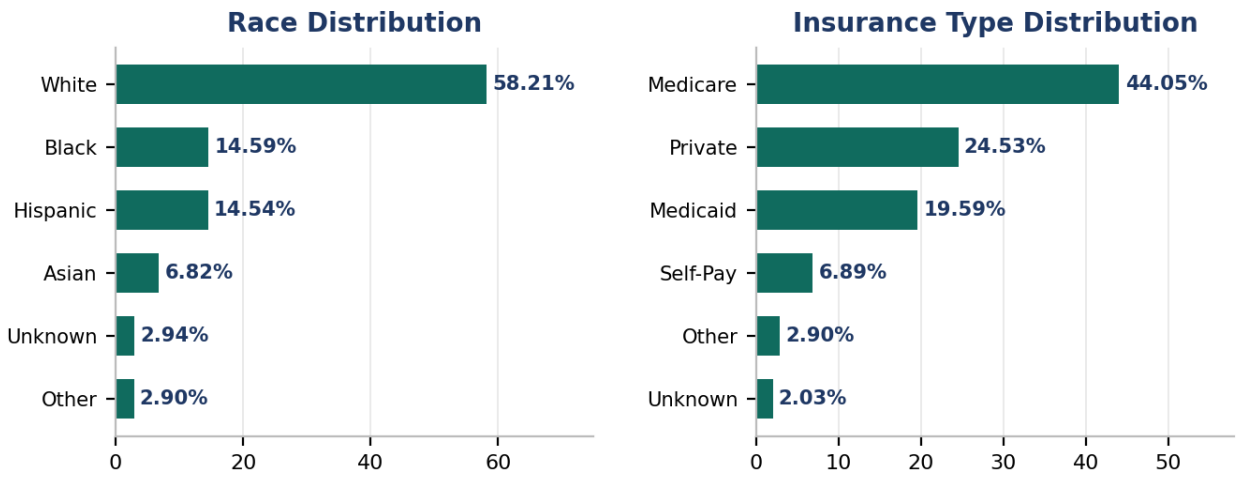


Figure 2. Dataset overview: race and insurance type.

White patients form the majority (58.21%), followed by Black (14.59%) and Hispanic (14.54%) patients. Medicare is the most common insurance type (44.05%), followed by private insurance (24.53%) and Medicaid (19.59%).

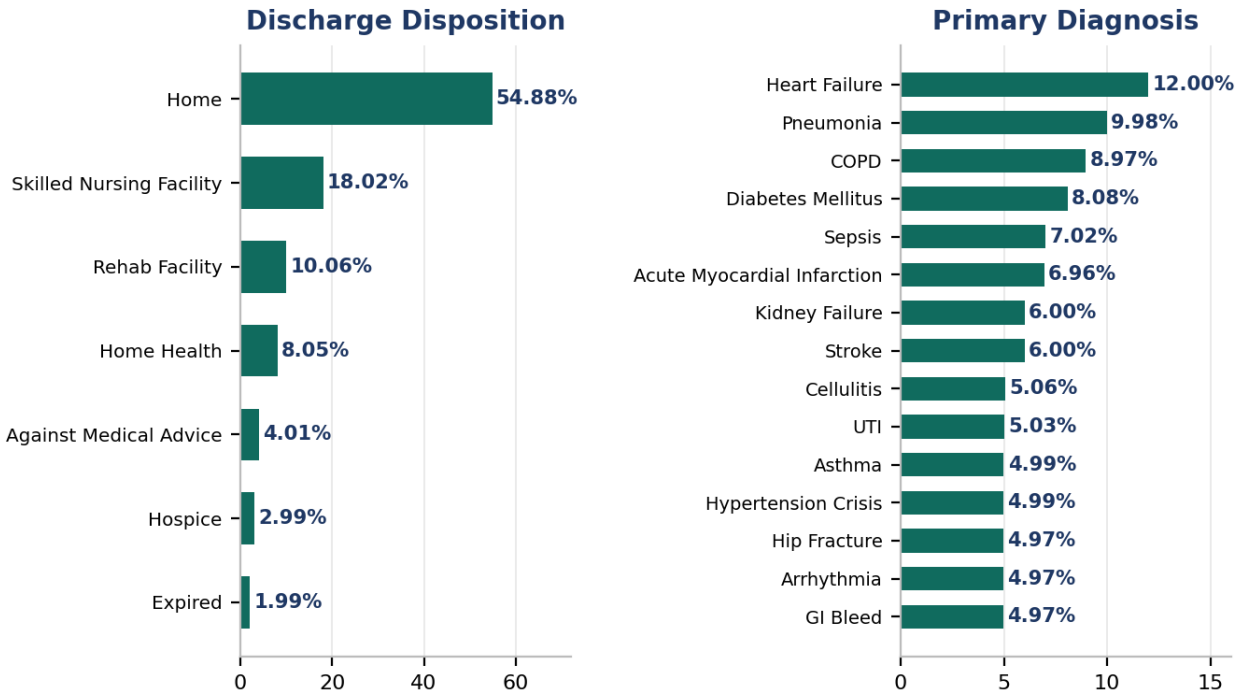


Figure 3. Dataset overview: discharge disposition and primary diagnosis.

More than half of the patients were discharged home (54.88%); 18.02% went to a skilled nursing facility. Heart failure was the most common diagnosis (12.00%), followed by pneumonia (9.98%) and COPD (8.97%); the remaining diagnoses each account for between 4.97% and 8.08% of patients.

Table 4. Descriptive statistics of the numerical variables (after imputation).

Variable	Mean	Median	Std	Min	Q1	Q3	Max	Skewness	Kurtosis
Age (years)	67.42	67.42	14.36	18	58	77	100	−0.09	−0.05
Length of Stay (days)	5.41	4	5.92	1	2	7	119	7.51	102.74
Previous Admissions	1.20	1	1.10	0	0	2	10	0.91	0.79
Number of Medications	10.00	10	3.16	0	8	12	28	0.25	0.02
Comorbidity Count	2.50	2	1.58	0	1	3	12	0.63	0.38
Haemoglobin (g/dL)	12.50	12.50	1.88	5	11.40	13.60	20	0.00	0.39
Creatinine (mg/dL)	1.01	0.90	0.52	0.30	0.67	1.21	7.79	1.90	6.68

- **Age:** mean and median are both 67.42 years and skewness is close to zero, so the distribution is nearly symmetric. The mean and median are identical partly because missing ages were replaced with the mean.
- **Length of stay:** strongly right-skewed (skewness 7.51, kurtosis 102.74). Most stays are short (median 4 days) but a few very long stays (up to 119 days) pull the mean up to 5.41 days.
- **Creatinine:** also right-skewed (skewness 1.90), with a median of 0.90 mg/dL below the mean of 1.01 mg/dL.
- **Medications, haemoglobin, comorbidity count and previous admissions:** roughly symmetric or mildly right-skewed, with no extreme values that suggest errors.

Results

Section 1: Age vs. 30-Day Readmission

Descriptive Analysis

Age Group (years)	Readmitted	Not Readmitted	Sample Size (n)
18–24	18.34%	81.66%	927
25–34	18.78%	81.22%	5,654
35–44	21.13%	78.87%	22,988
45–64	25.65%	74.35%	168,216
65+	31.83%	68.17%	302,215
Total	**29.08%**	**70.92%**	**500,000**

The table shows the percentage of patients readmitted and not readmitted within 30 days in each age group. Readmission rates increased steadily with age, from 18.34% among patients aged 18–24 to 31.83% among those aged 65 and above.

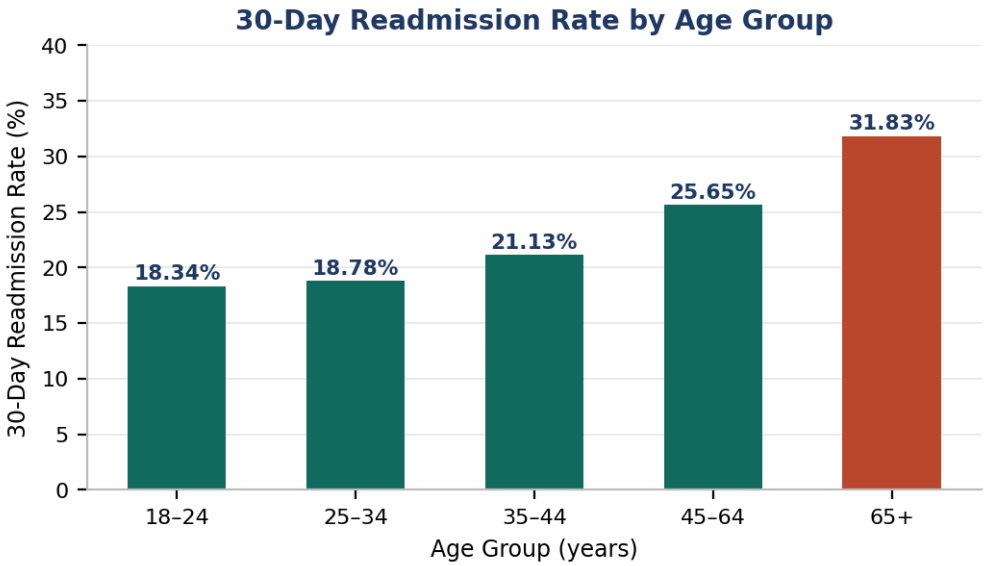


Figure 4. 30-day readmission rate by age group.

Chi-Square Test of Independence

Chi-square statistic: 3,112.82
Degrees of freedom: 4
p-value: < 0.001
Cramér's V: 0.0789 (weak association)

Since p < 0.001, the null hypothesis was rejected: age group and 30-day readmission are associated. However, Cramér's V (below 0.10) shows that the association is weak. Comparing each age group with the next younger group confirmed that readmission was significantly higher at every step from age 25–34 upward, while ages 18–24 and 25–34 did not differ significantly (Holm-adjusted p = 0.75).

Independent Two-Sample t-Test

Group	Average Age (years)	Sample Size (n)
Readmitted	69.59	145,413
Not readmitted	66.53	354,587
Mean difference (Readmitted – Not readmitted)	**3.06**	

Mean difference: 3.06
t-statistic: 68.69
Degrees of freedom: 499,998
p-value: < 0.001
Cohen's d: 0.214

Readmitted patients were approximately 3.06 years older on average than non-readmitted patients (69.59 vs 66.53 years). Cohen's d = 0.214 indicates a small effect. Welch's t-test, which does not assume equal variances, gave the same result (t = 69.01, p < 0.001).

Note: 29,965 patients had a missing age and were assigned the mean age (67.42 years), so they fall in the 65+ group. A check using only patients with a recorded age gave the same conclusion (see Limitations).

> Conclusion: Age was significantly associated with 30-day readmission in both the age-group analysis and the comparison of average ages. Readmission increased steadily with age, but the strength of the association was weak to small, so age alone does not fully explain readmission. This is an association, not proof of causation.

Section 2: Gender vs. 30-Day Readmission

Descriptive Analysis

Gender	Readmitted	Not Readmitted	Sample Size (n)
Female	29.11%	70.89%	244,943
Male	29.02%	70.98%	230,053
Other	29.79%	70.21%	4,887
Unknown	29.21%	70.79%	20,117
Total	**29.08%**	**70.92%**	**500,000**

Readmission rates were very similar across all four gender categories, ranging from 29.02% to 29.79%.

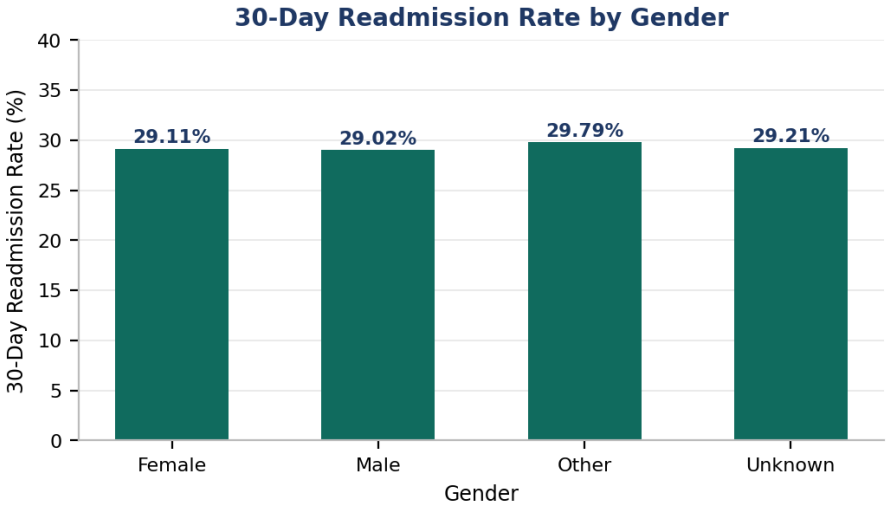


Figure 5. 30-day readmission rate by gender.

Chi-Square Test of Independence

Chi-square statistic: 1.84

Degrees of freedom: 3

p-value: 0.6057

Cramér's V: 0.0019 (negligible association)

Since p = 0.6057 is greater than 0.05, we failed to reject the null hypothesis. There is insufficient evidence of an association between gender and 30-day readmission, and Cramér's V shows that any association is negligible.

> Conclusion: Gender was not significantly associated with 30-day readmission ($\chi^2 = 1.84$, $p = 0.6057$). Other patient characteristics are likely to have a greater influence on readmission risk.

Section 3: Race vs. 30-Day Readmission

Descriptive Analysis

Race	Readmitted	Not Readmitted	Sample Size (n)
Asian	28.76%	71.24%	34,084
Black	28.95%	71.05%	72,965
Hispanic	29.14%	70.86%	72,708
Other	28.43%	71.57%	14,485
Unknown	28.64%	71.36%	14,716
White	29.19%	70.81%	291,042
Total	**29.08%**	**70.92%**	**500,000**

Readmission rates were relatively similar across all six race categories, ranging from 28.43% to 29.19%.

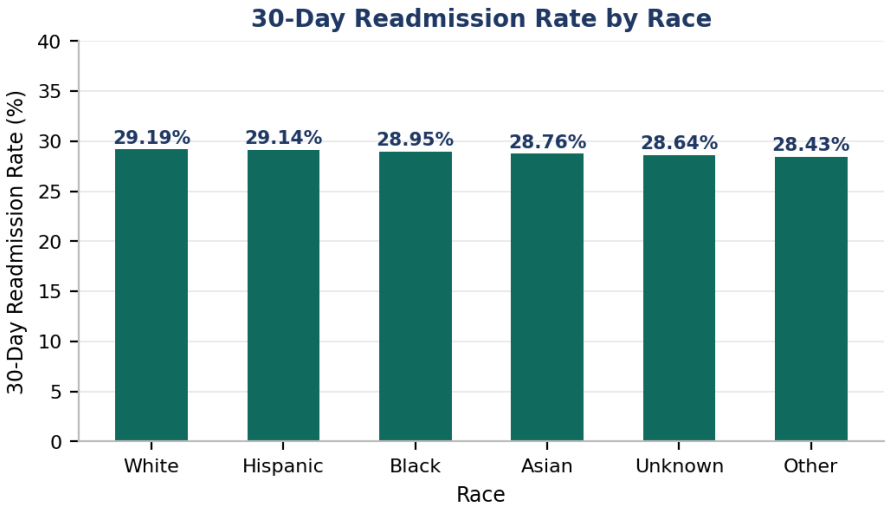


Figure 6. 30-day readmission rate by race.

Chi-Square Test of Independence

Chi-square statistic: 8.47
Degrees of freedom: 5
p-value: 0.1322
Cramér's V: 0.0041 (negligible association)

Since p = 0.1322 is greater than 0.05, we failed to reject the null hypothesis. There is insufficient evidence of an association between race and 30-day readmission, and Cramér's V shows that any association is negligible.

> Conclusion: Race was not meaningfully associated with 30-day readmission in this dataset. Readmission rates were similar in every racial group.

Section 4: Admission Type vs. 30-Day Readmission

Descriptive Analysis

Admission Type	Readmitted	Not Readmitted	Sample Size (n)
Elective	28.03%	71.97%	100,025
Emergency	29.79%	70.21%	275,277
Urgent	28.35%	71.65%	124,698
Total	**29.08%**	**70.92%**	**500,000**

Emergency admissions had the highest readmission rate (29.79%), followed by urgent (28.35%) and elective (28.03%) admissions.

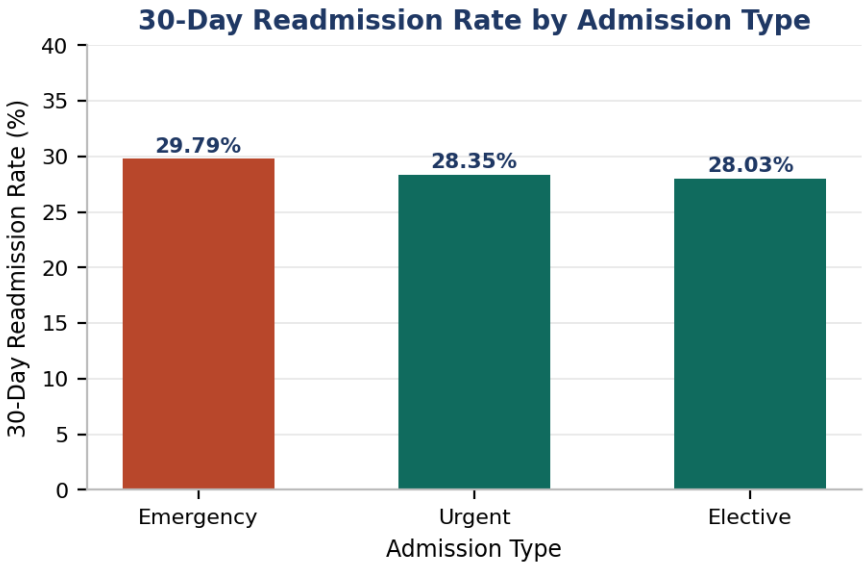


Figure 7. 30-day readmission rate by admission type.

Chi-Square Test of Independence

Chi-square statistic: 153.17

Degrees of freedom: 2

p-value: < 0.001

Cramér's V: 0.0175 (negligible association)

Since p < 0.001, the null hypothesis was rejected: admission type and 30-day readmission are associated. However, Cramér's V shows that the association is negligible, so admission type alone has very little influence on readmission.

> Conclusion: Admission type was statistically associated with 30-day readmission, with emergency admissions highest, but the association was negligible in strength. Other factors are likely to matter more.

Section 5: Length of Stay vs. 30-Day Readmission

Descriptive Analysis

Group	Average LOS (days)	Sample Size (n)
Readmitted	5.93	145,413
Not readmitted	5.20	354,587
Mean difference (Readmitted – Not readmitted)	**0.73**	

Patients who were readmitted had a longer initial hospital stay (5.93 days) than those who were not (5.20 days), a difference of 0.73 days.

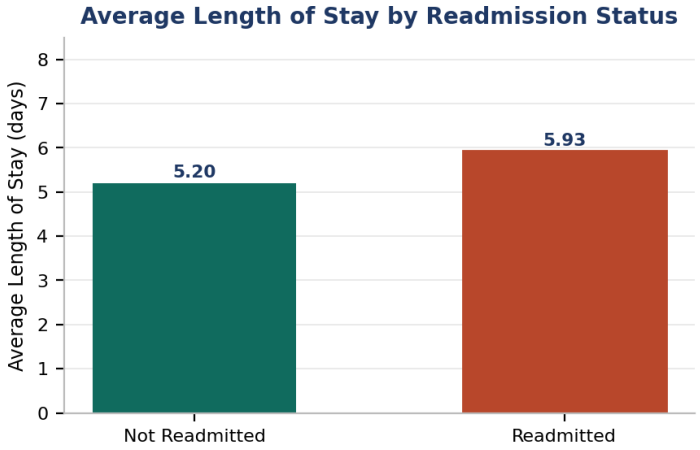


Figure 8. Average length of stay for readmitted and non-readmitted patients.

Independent Two-Sample t-Test

Mean difference: 0.73
t-statistic: 39.58
Degrees of freedom: 499,998
p-value: < 0.001
Cohen's d: 0.123

Since p < 0.001, the null hypothesis was rejected. Cohen's d = 0.123 is a very small effect (below the 0.2 level usually regarded as small). Welch's t-test gave the same conclusion (t = 33.60).

Non-Parametric Confirmation: Mann–Whitney U Test

Length of stay is strongly skewed (skewness 7.51, kurtosis 102.74), so the Mann–Whitney U test was performed as a check. It asks whether readmitted patients tend to have longer stays than non-readmitted patients.

Mann–Whitney U statistic: 27,000,486,579
p-value: 1.2828×10^{-154}

The Mann–Whitney U test also showed a statistically significant difference, so both tests agree. The difference partly reflects a small number of very long stays: when stays longer than 30 days are excluded, the difference falls to 0.36 days (Cohen's d = 0.087) but remains statistically significant.

> Conclusion: Length of stay was significantly associated with 30-day readmission: readmitted patients stayed about 0.7 days longer on average. The effect size is very small, and the findings do not establish that a longer stay causes readmission.

Section 6: Primary Diagnosis vs. 30-Day Readmission

Descriptive Analysis

Primary Diagnosis	Readmitted	Not Readmitted	Sample Size (n)
COPD	32.48%	67.52%	44,826
Sepsis	32.38%	67.62%	35,103
Acute Myocardial Infarction	32.31%	67.69%	34,780
Heart Failure	32.06%	67.94%	59,986
Asthma	27.86%	72.14%	24,967
Hypertension Crisis	27.81%	72.19%	24,940
Hip Fracture	27.54%	72.46%	24,869
GI Bleed	27.53%	72.47%	24,853
Pneumonia	27.39%	72.61%	49,912
UTI	27.33%	72.67%	25,154
Kidney Failure	27.31%	72.69%	30,020
Cellulitis	27.19%	72.81%	25,315
Stroke	27.13%	72.87%	30,017
Arrhythmia	27.11%	72.89%	24,869
Diabetes Mellitus	27.05%	72.95%	40,389
Total	**29.08%**	**70.92%**	**500,000**

COPD had the highest readmission rate (32.48%), followed by sepsis (32.38%), acute myocardial infarction (32.31%) and heart failure (32.06%). Diabetes mellitus had the lowest rate (27.05%). The difference between the highest and lowest rates was approximately 5.44 percentage points.

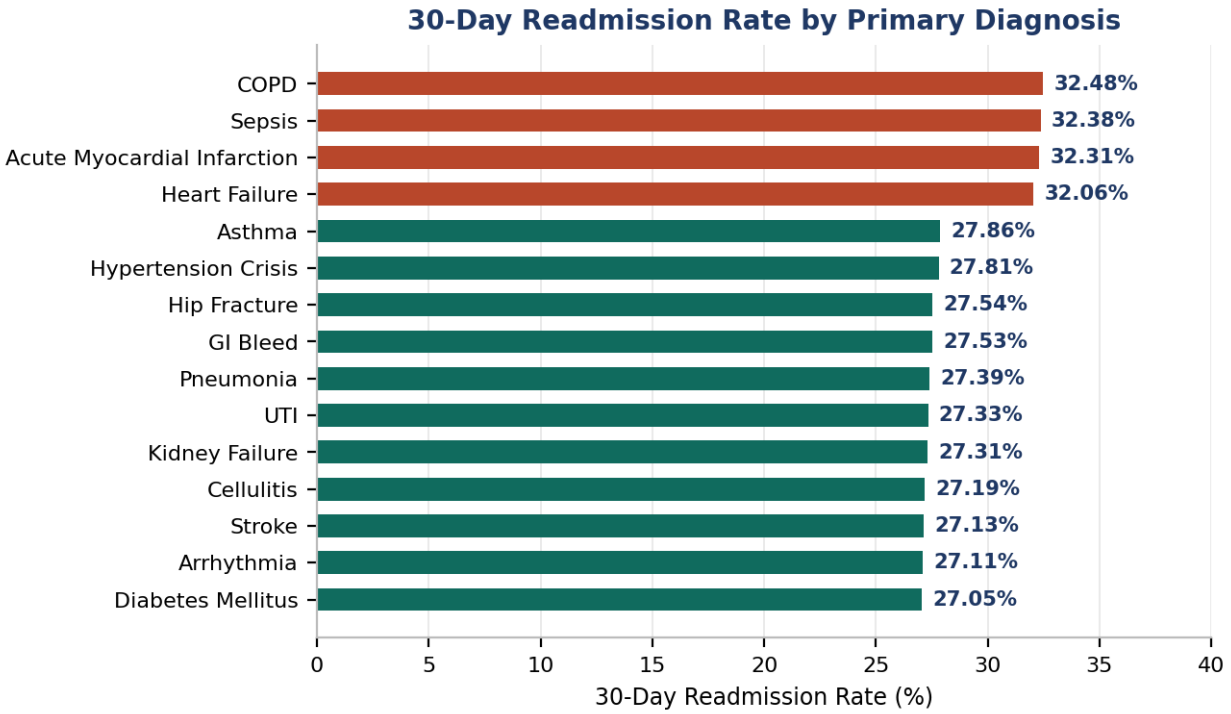


Figure 9. 30-day readmission rate by primary diagnosis (four highest-rate diagnoses highlighted).

Chi-Square Test of Independence

Chi-square statistic: 1,344.33

Degrees of freedom: 14

p-value: 1.5604×10^{-278} (< 0.001)

Cramér's V: 0.0519 (weak association)

Since p < 0.001, the null hypothesis was rejected: primary diagnosis and 30-day readmission are associated, although Cramér's V shows the association is weak. Pairwise comparisons (Holm-adjusted) showed that each of the four highest-rate diagnoses differs significantly from each of the other eleven diagnoses (32.28% vs 27.37% overall, a gap of 4.92 points), while no two diagnoses within the same group differ significantly.

> Conclusion: Primary diagnosis was statistically associated with 30-day readmission. COPD, sepsis, acute myocardial infarction and heart failure had the highest readmission rates. These findings may help identify diagnosis groups for further investigation, but they do not establish that a diagnosis causes readmission.

Section 7: Logistic Regression for Previous Admissions, Medications and Comorbidity Count

Descriptive Analysis

The three count variables were first compared with readmission rates. Readmission increased steadily with both the number of previous admissions and the comorbidity count.

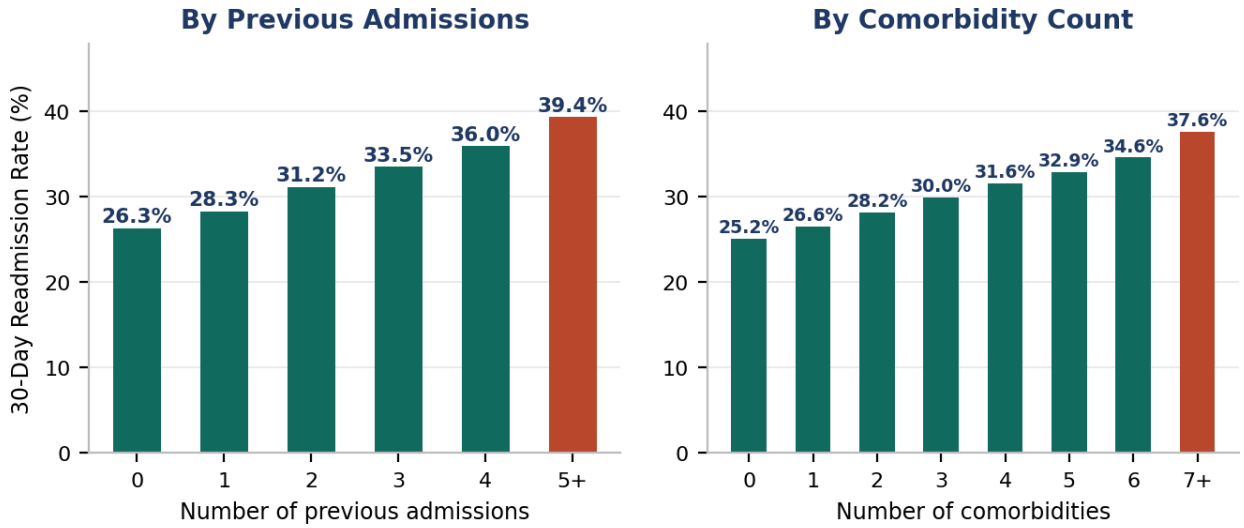


Figure 10. 30-day readmission rate by number of previous admissions and by comorbidity count (5+ and 7+ combine all higher values).

Logistic Regression Analysis

- **Dependent variable:** Readmitted_30_Days (Yes = 1, No = 0)
- **Independent variables:** number of previous admissions, number of medications, comorbidity count
- **Sample size:** 500,000 patients

Variable	Coefficient	Odds Ratio	95% CI	p-value
Previous admissions	0.1170	1.1241	1.1180 – 1.1303	< 0.001
Number of medications	0.0034	1.0034	1.0014 – 1.0053	< 0.001
Comorbidity count	0.0789	1.0821	1.0780 – 1.0863	< 0.001

All three p-values are below 0.05, so each variable is significantly associated with 30-day readmission after accounting for the other two.

Interpretation of Results

- **Previous admissions:** each additional previous admission was associated with 12.4% higher odds of 30-day readmission, holding the other two variables constant.
- **Number of medications:** each additional medication was associated with approximately 0.34% higher odds of readmission. This association is very small.
- **Comorbidity count:** each additional coexisting condition was associated with approximately 8.2% higher odds of readmission.

Odds ratios describe changes in odds per one additional unit; they are not changes in probability, and because each variable is measured in different units they should not be ranked against each other. They show association, not causation.

Model Strength

McFadden's pseudo-R^2: 0.00557 (about 0.56%)
Likelihood-ratio test: χ^2 = 3,358.1, df = 3, p < 0.001

The model is statistically better than a model with no predictors, but the low pseudo-R^2 shows that these three variables explain only a small part of who is readmitted.

Model Evaluation

To check how well the model predicts patients it has not seen, it was fitted on 400,000 randomly selected patients (80%) and tested on the remaining 100,000 patients (20%), with the same readmission rate (29.08%) in both sets. Patients were flagged as high risk when their predicted risk was at least the average readmission rate (29.1%).

Measure	Result	Meaning
AUC	0.550 (95% CI 0.546–0.554)	0.50 = no better than chance; 1.00 = perfect
Accuracy	54.8%	Share of all patients classified correctly
Sensitivity	50.3%	Share of readmitted patients correctly flagged
Specificity	56.6%	Share of non-readmitted patients correctly not flagged
Precision	32.2%	Share of flagged patients who were readmitted (overall rate 29.1%)
Brier score	0.2048	Prediction error; reference model 0.2062 (lower is better)
Calibration slope	1.04	1.0 = predicted risks match observed rates

Confusion matrix (test set)

	Flagged High Risk	Not Flagged	Total
Readmitted	14,627	14,456	29,083
Not readmitted	30,755	40,162	70,917
Total	**45,382**	**54,618**	**100,000**

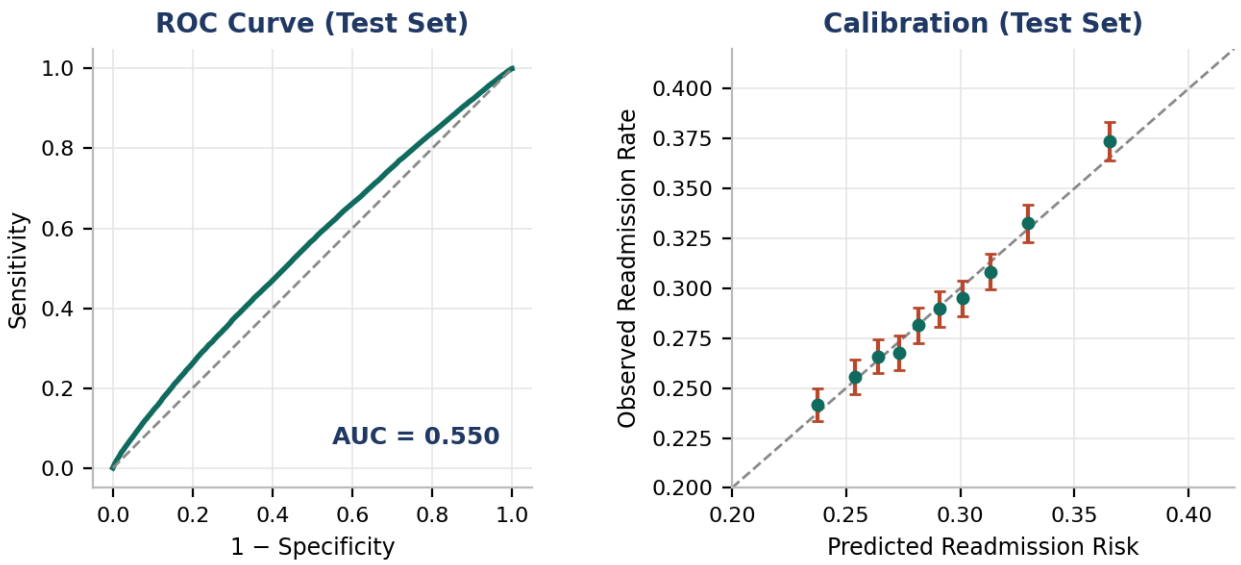


Figure 11. ROC curve and calibration plot of the logistic regression model on the test set.

An AUC of 0.55 is only slightly better than chance (0.50), so the model has limited ability to tell which patients will be readmitted. At the chosen threshold it flagged about half of the readmitted patients (50.3%), but only 32.2% of the patients it flagged were actually readmitted, compared with 29.1% overall. The predicted risks did match the observed rates well (calibration slope 1.04), so the model is consistent but not a strong predictor. Repeating the split with five other random seeds gave similar AUCs (0.547–0.551).

Conclusion: Previous admissions, number of medications and comorbidity count were each significantly associated with higher odds of 30-day readmission, with previous admissions and comorbidity count showing the clearest effects. However, the model has limited predictive ability on unseen data (AUC = 0.55), so these findings describe associations rather than a tool for predicting individual patients. Because the dataset is synthetic, the results describe patterns within this dataset and are not clinical evidence.

Section 8: Discharge Disposition vs. 30-Day Readmission

Descriptive Analysis

Discharge Disposition	Readmitted	Not Readmitted	Sample Size (n)
Against Medical Advice	34.97%	65.03%	20,038
Skilled Nursing Facility	30.18%	69.82%	90,110
Expired	29.02%	70.98%	9,936
Rehab Facility	28.58%	71.42%	50,292
Home Health	28.53%	71.47%	40,253
Home	28.52%	71.48%	274,416
Hospice	28.08%	71.92%	14,955
Total	**29.08%**	**70.92%**	**500,000**

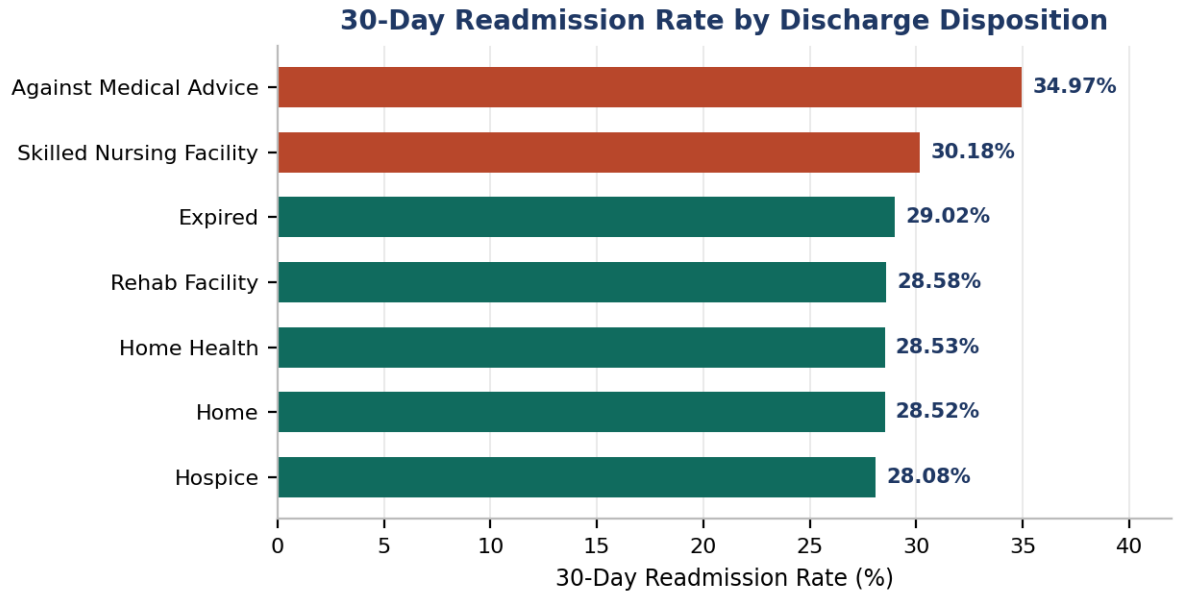


Figure 12. 30-day readmission rate by discharge disposition.

Chi-Square Test of Independence

Chi-square statistic: 450.44
Degrees of freedom: 6
p-value: < 0.001
Cramér's V: 0.0300 (weak association)

- Discharge against medical advice had the highest readmission rate (34.97%).
- Discharge to a skilled nursing facility had the second-highest rate (30.18%).
- Hospice had the lowest rate (28.08%), and the difference between the highest and lowest rates was approximately 6.89 percentage points.

Since p < 0.001, the null hypothesis was rejected, although Cramér's V shows the association is weak. Comparing each discharge route with discharge home (Holm-adjusted) showed that only discharge against medical advice (+6.5 percentage points) and discharge to a skilled nursing facility (+1.7 points)

differed significantly from discharge home.

Note: 9,936 patients are recorded as “Expired” (died in hospital) yet 2,883 of them (29.02%) carry a readmission outcome. This is a quirk of the synthetic data. Excluding them did not change the conclusions (see Limitations).

> Conclusion: Discharge disposition was statistically associated with 30-day readmission. Patients discharged against medical advice had the highest rate, and those discharged to skilled nursing facilities also had a higher rate. The overall association was weak, and the findings do not establish that the discharge route caused readmission.

Section 9: Haemoglobin Level vs. 30-Day Readmission

Descriptive Analysis

Group	Average Haemoglobin (g/dL)	Sample Size (n)
Readmitted	12.49	145,413
Not readmitted	12.50	354,587
Mean difference (Readmitted − Not readmitted)	**−0.01**	

The average haemoglobin level was 12.49 g/dL in readmitted patients and 12.50 g/dL in non-readmitted patients, an almost identical result.

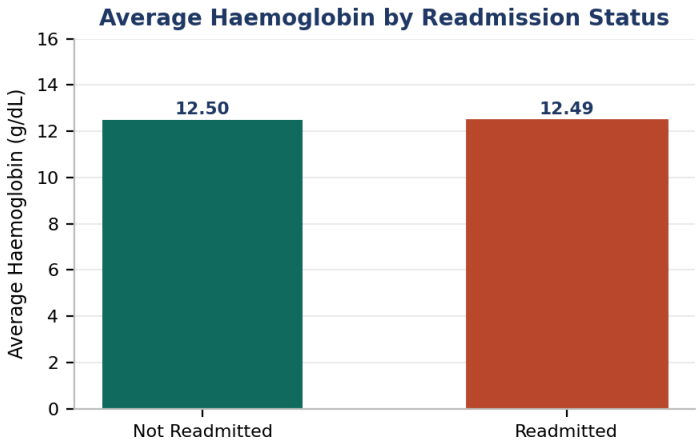


Figure 13. Average haemoglobin level for readmitted and non-readmitted patients.

Independent Two-Sample t-Test

Mean difference: −0.008
t-statistic: −1.34
Degrees of freedom: 499,998
p-value: 0.1802
Cohen's d: −0.0042

Since p = 0.1802 is greater than 0.05, we failed to reject the null hypothesis. There is insufficient evidence of a difference in haemoglobin level between the two groups. Cohen's d shows that the difference is negligible, and Welch's t-test gave the same result (t = −1.34).

Conclusion: No significant difference in haemoglobin level was found between readmitted and non-readmitted patients. This describes this synthetic dataset and does not show that haemoglobin has no relationship with readmission in real patients.

Section 10: Creatinine Level vs. 30-Day Readmission

Descriptive Analysis

Group	Average Creatinine (mg/dL)	Sample Size (n)
Readmitted	1.0123	145,413
Not readmitted	1.0127	354,587
Mean difference (Readmitted − Not readmitted)	**−0.0003**	

The average creatinine level was 1.0123 mg/dL in readmitted patients and 1.0127 mg/dL in non-readmitted patients, an almost identical result.

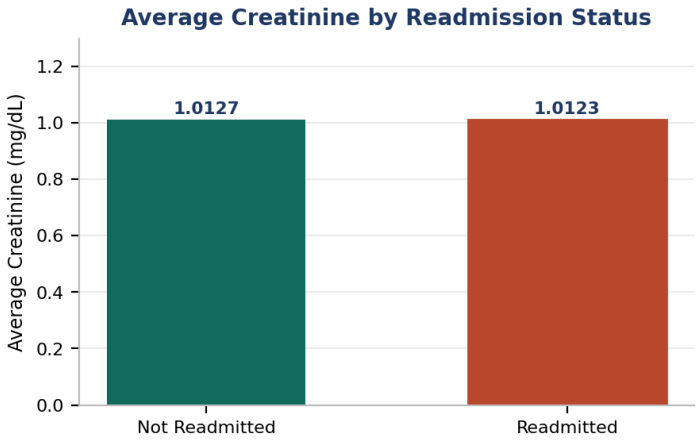


Figure 14. Average creatinine level for readmitted and non-readmitted patients.

Independent Two-Sample t-Test

Mean difference: −0.0003
t-statistic: −0.196
Degrees of freedom: 499,998
p-value: 0.8446
Cohen's d: −0.0006

Since p = 0.8446 is greater than 0.05, we failed to reject the null hypothesis. There is insufficient evidence of a difference in creatinine level between the two groups, and the effect size is negligible.

Conclusion: No significant difference in creatinine level was found between readmitted and non-readmitted patients. This describes this synthetic dataset and does not show that creatinine has no relationship with readmission in real patients.

Section 11: Insurance Type vs. 30-Day Readmission

Descriptive Analysis

Insurance Type	Readmitted	Not Readmitted	Sample Size (n)
Medicaid	29.17%	70.83%	97,925
Medicare	29.06%	70.94%	220,270
Other	29.31%	70.69%	14,511
Private	29.13%	70.87%	122,661
Self-Pay	28.66%	71.34%	34,465
Unknown	29.26%	70.74%	10,168
Total	**29.08%**	**70.92%**	**500,000**

Readmission rates were very similar across all six insurance categories, from 28.66% for Self-Pay patients to 29.31% for patients with Other insurance, a difference of only 0.65 percentage points.

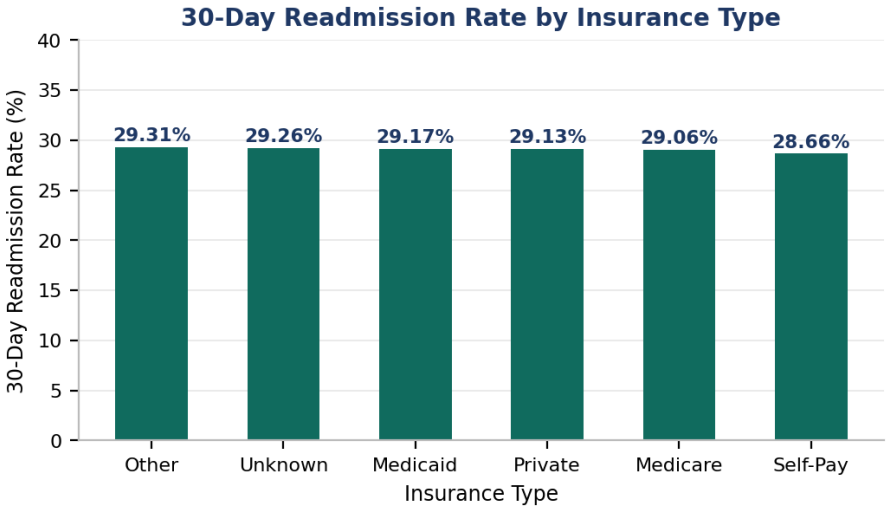


Figure 15. 30-day readmission rate by insurance type.

Chi-Square Test of Independence

Chi-square statistic: 4.05

Degrees of freedom: 5

p-value: 0.5428

Cramér's V: 0.0028 (negligible association)

Since p = 0.5428 is greater than 0.05, we failed to reject the null hypothesis. There is insufficient evidence of an association between insurance type and 30-day readmission.

> Conclusion: Insurance type was not associated with 30-day readmission. Patients across the six insurance categories had very similar readmission rates, and the measured association was negligible.

Limitations

- **Synthetic data.** The dataset was generated with Claude AI and contains no real patients. The findings describe patterns within this dataset and are not clinical evidence for any hospital.
- **Association, not causation.** Each factor was tested one at a time, so overlapping effects (for example, older patients tending to have more conditions) were not separated.
- **Statistical versus practical significance.** With 500,000 patients, even tiny differences are statistically significant. Most effect sizes were small (largest Cramér's V = 0.08; largest Cohen's d = 0.21), so effect sizes should be read together with p-values.
- **Imputation.** Missing values were imputed, and the 29,965 imputed ages all fall in the 65+ group. Using only patients with a recorded age, the age results were unchanged (65+ readmission 32.15% vs 31.83%).
- **Data quirks.** 9,936 patients recorded as “Expired” carry a readmission outcome, and age values pile up at 100 and creatinine values at 0.30. Excluding the “Expired” patients did not change the conclusions.
- **Limited prediction.** The logistic regression model used only three variables and showed limited predictive ability (AUC = 0.55). It was tested on a hold-out sample from the same dataset, not on data from another hospital.

Recommendations

Because the data are synthetic, the recommendations below are suggestions for further investigation rather than established clinical guidance.

1. Prioritize discharge planning and follow-up for older patients (aged 65 and above), as age showed the strongest association with 30-day readmission.
2. Flag patients with two or more previous admissions or several comorbidities for case management, as both showed a clear, graded increase in readmission.
3. Develop diagnosis-specific discharge pathways for COPD, sepsis, acute myocardial infarction and heart failure, which had the highest readmission rates.
4. Strengthen hand-over and early follow-up for patients discharged against medical advice or to skilled nursing facilities.
5. Review discharge-readiness criteria for patients with long hospital stays and monitor emergency admissions, although both effects were small.
6. Do not use gender, race, insurance type, haemoglobin or creatinine to decide who receives extra support, as none showed a meaningful association. Continue to monitor for fairness.
7. Build a routine dashboard to monitor 30-day readmission rates by age group, diagnosis and discharge route.

Next Steps

This project focused mainly on single-factor statistical analysis and a three-variable logistic regression. Future work could extend the analysis by:

1. Performing multiple logistic regression with all factors to identify the independent effect of each factor while controlling for the others.
2. Conducting further pairwise (post-hoc) comparisons for the remaining groups, such as admission type and insurance type.
3. Building an interactive Power BI or Excel dashboard for hospital administrators, and including additional variables such as laboratory results, medication adherence and follow-up attendance.
4. Applying machine learning models (for example random forest) to predict readmission risk and comparing them with the logistic regression.
5. Validating the findings using real hospital data from multiple facilities, including Nigerian hospitals and teaching hospitals, to improve generalizability.